

Mathematical Methods in Uncertainty Quantification (MMUQ) – L03

Faizan Anwar (faizan.anwar@tum.de)

Technical University of Munich

School of Engineering and Design

Chair of Hydrology and River Basin Management (Prof. Disse)

MMUQ L03 – 2nd Assignment and Local Sensitivity Analysis

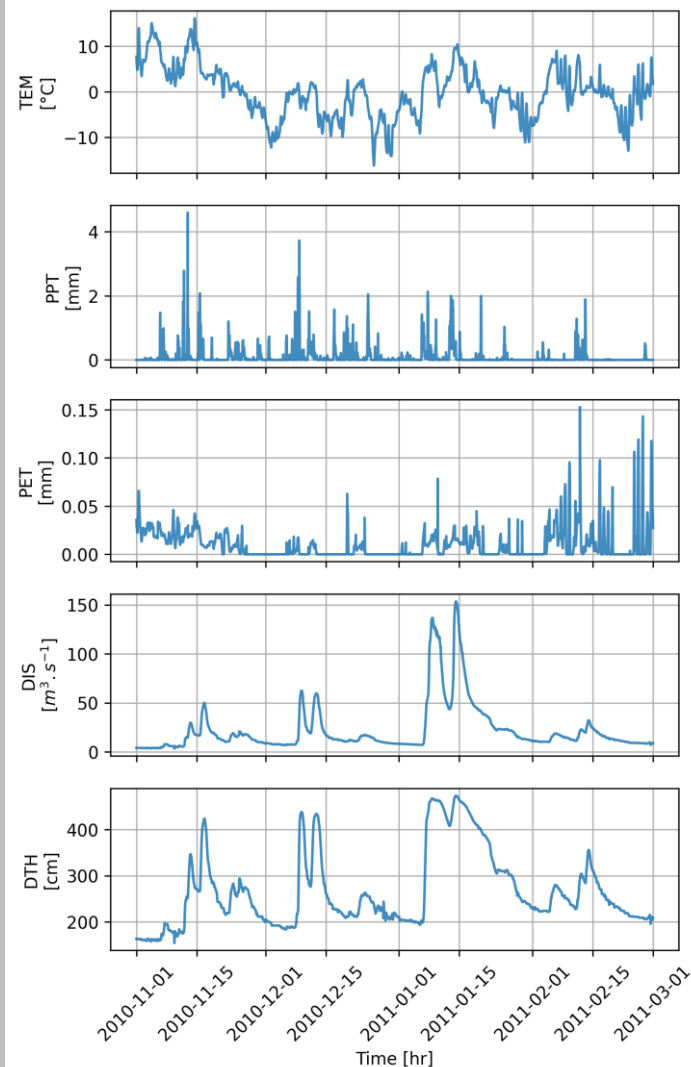
- Assignment
- Tasks
- Local Sensitivity Analysis

Assignment – 2

- Local sensitivity analysis (SA) of model parameters
- Effects of each parameter on the OF
 - Relative perturbation
 - Quantity of influence on model accuracy
- Use of a calibrated model
- Reference: Uncertainty quantification, Smith 2014, Book

Assignment – 2, Problem

- Given
 - Data from ASGT-1
 - Optimized parameters (Xs)
- Which X(s) affect the OFV (Y) the most
 - Which have no effect
 - Explanation(s)
- Investigate
 - X and Y linkage
- Relative changes!
 - % change in X leads to % change in Y
- Suggestions on how to have a non-sensitive X



Assignment – 2, Tasks

- Loop through each optimized Parameter (X)
- Apply relative changes
 - say $\pm 20\%$
- Record changes in OFV (Y)
- Plot relative X and Y change scatters
- Rank most to least sensitive
- Explain why given X is sensitive or not
- Suggest how to make X not sensitive
- Identify X(s) that are not important

Group Work

That was it for today, thanks!